The Reuse Decision Framework

The outcome of both summer directed studies, built from observation on site and the online research database. It does not give a material a score; it asks whether the evidence is enough for one specific use, then what to do next.

Can this particular component perform this particular role in this particular building?

Suitability cannot be assigned before the next use is known.

1 Identify

the material reuse passport: a record that enters the decision

Material record: what is known

Material ID	unique reference
Component type	beam, joist, stud, window, flooring, brick, cladding, fixture
Source	donor building and original location
Original use	previous architectural role
Material	species, composition or product type
Age	known or estimated
Dimensions	measured size and variation
Quantity	available stock
Physical condition	damage, deterioration, previous alterations
Treatment	known treatment, coating or contamination
Recovery status	in place, recovered, stored, processed, allocated
Documentation	photographs, scans, drawings, tests, records

Application record: what a proposed use needs

Proposed use	the architectural application being tested
Building system	structure, envelope, interior, service or site
Required performance	the relevant technical requirements
Evidence available	inspection, grading, testing, certification
Evidence missing	information still required
Processing required	clean, repair, cut, grade, modify, test
Connection strategy	relationship to the proposed assembly
Regulation	requirements and professional review
Environmental value	avoided new material, processing, transport
Labour and logistics	recovery, storage, fabrication, construction
Architectural value	dimensions, patina, history, irregularity
Future circularity	reversibility, replaceability, future recovery
Evidence confidence	high, medium, low or unresolved
Decision status	one of the eight decisions in stage 3

One component, many application records: if a timber member lacks the evidence for structural reuse, the same record can be tested for secondary framing, a partition, a screen or furniture. Unknowns are recorded, never hidden; they set how much more investigation is needed.

2 Evaluate

twelve criteria, A to L; their weight changes with material, use, evidence and consequence of failure

Three questions come first

A Material identity
What is the component?

- Material type
- Material subtype
- Original product or component
- Original application
- Age
- Source building
- Provenance
- Known or unknown composition
- Treatment or coating
- Potential contamination

H Environmental value
What value is kept by keeping it in circulation?

- Avoided disposal
- Avoided production of virgin material
- Embodied carbon implications
- Transportation impact
- Processing impact
- Waste generated through recovery
- Comparison with an equivalent new material
- Potential for another future use

E Deconstruction and recovery
Can it come out in a form that stays useful?

- Accessibility
- Existing connection type
- Reversibility
- Difficulty of removal
- Risk of damage during removal
- Labour required
- Equipment required
- Recoverable yield
- Ability to separate it from adjacent work

Then nine more questions

B Physical condition
Damage, weathering, decay, corrosion, cracking, deformation, moisture, biological damage, old fasteners, remaining service life

C Geometry and quantity
Dimensions, thickness, section, length, shape, irregularity, quantity, repeatability, tolerances

D Technical performance
Structural, enclosure, thermal, moisture, acoustic, fire, durability, weathering, finish, maintenance

F Processing
Cleaning, de-nailing, trimming, cutting, resurfacing, grading, testing, machining, repair, reinforcement

J Regulation and risk
Building code, testing, certification, hazardous materials, liability, traceability, documentation

G Logistics
Location, transport, storage, handling, timing, availability, sequencing with the construction schedule

I Economics and labour
Recovery, processing, testing, transport, storage, fabrication and design labour, against new material

K Architectural value
Patina, visible history, cultural significance, dimensional limits, design opportunity, fit with the assembly

L Future circularity
Reversibility, adaptable connections, future disassembly, identity, documentation for later users

Five levels of assessment

· evidence confidence is recorded at every level: high, medium, low or unresolved

1 Rapid screening
worth protecting during deconstruction?

2 Documentation
identity, size, quantity, condition, provenance

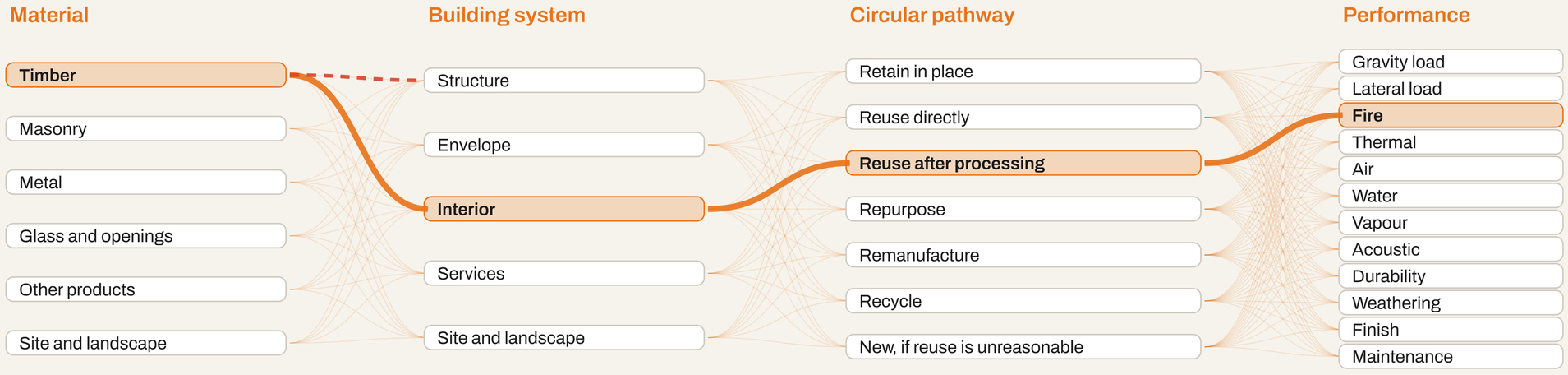
3 Technical assessment
testing, grading, specialist assessment

4 Design compatibility
against a role, connection and assembly

5 Holistic decision
environment, cost, logistics, design, future

3 Decide

match the part to a role, weigh evidence against consequence, then classify



Example. A recovered joist lacks grading for structure (dashed), so it is tested again as an interior partition, reused after processing, against fire and finish.

Evidence against the consequence of failure

	limited evidence	moderate evidence	strong evidence
low consequence	Potential candidate	Reuse generally credible	Reuse credible
moderate consequence	Further verification required	Conditional reuse	Reuse subject to project requirements
high consequence	Do not proceed	Testing or professional assessment required	Potentially suitable, subject to code and review

Eight decisions, each with the architect's next move

Reuse-ready	adequate evidence, feasible recovery, a defined use	**integrate into design**
Conditional reuse	promising, with unresolved questions	**test, document or prototype**
Reuse after processing	viable after a defined intervention	**add processing to the workflow**
Redirect to another use	this use is wrong, another remains viable	**match to a more compatible role**
Design-dependent	depends on changing assembly, geometry, joint	**redesign around the stock**
Reserve	viable, but no receiving use yet	**catalogue and store**
Recycle	the material is recoverable, the component is not	**keep apart from reuse stock**
Exclude	hazard, poor performance or destructive recovery	**treat or dispose appropriately**

4 Design

take the least transformative path that credibly meets the use

outputs: inventory · passports · use-matching · uncertainty log · decision tree · flow map · assembly atlas · prototype

Direct reuse	original component and geometry
Preparation	substantially unchanged
Repair	most of the component
Modification	identity kept, geometry changes
Testing or regrading	geometry kept, confidence rises
Remanufacture	material kept, form partly lost
Recycle	material kept, identity lost
Exclude	no reuse value kept

